

AKHILESH KUMAR MISHRA

SOFTWARE ENGINEER / FULL STACK ENGINEER

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PROFILE

Versatile Software Development Engineer with extensive experience building secure, scalable full-stack web applications and microservice architectures across multiple technology stacks (MERN, Java Full Stack, Python Full Stack). Expert in system design, API development, and database engineering. Proven success in re-architecting monolithic services into containerized microservices, building real-time telemetry pipelines via WebSockets, and optimizing transaction-heavy ledger platforms. Led development teams and streamlined DevOps workflows using Docker and CI/CD.

CORE TECHNOLOGY

LANGUAGES

Python, Java, JavaScript (ES6+), TypeScript, SQL, C++, HTML5, CSS3

FRAMEWORKS

Spring Boot, Django, FastAPI, Node.js, Express.js, React.js, Next.js, Redux

DATA & MESSAGING

MySQL, PostgreSQL, MongoDB, Redis, Apache Kafka, Mongoose, Hibernate, JPA

DEVOPS & TOOLS

Docker, Git, GitHub Actions, CI/CD, Postman, Linux, OpenCV, Pandas, Maven

EXPERIENCE

12/2023 – PRESENT

New Delhi, India

Software Development Engineer

CSC e-Governance Services India Ltd. (MeitY, Government of India)

Ministry of Electronics & Information Technology, Govt. of India

- Re-architected monolithic backend services into distributed microservices using **Spring Boot**, **FastAPI**, and **Node.js**, improving performance for **2 million+ users**.
- Architected transactional endpoints for AEPS, MATM, and UPI platforms, managing **500,000+ daily transactions** with an **~80% success rate**.
- Secured backend APIs utilizing **JWT Authentication** and **RBAC**, integrating device binding and OTP security, reducing unauthorized access attempts by 99.9%.
- Architected and optimized databases using **MySQL**, **PostgreSQL**, and **MongoDB**, utilizing sharding and query tuning to reduce read/write latency by 45%.
- Established event-driven communication using **Kafka**, **Celery**, and **Redis** cache, boosting message processing throughput by 35%.
- Integrated transaction monitoring dashboards using **React.js**, **Redux**, and **WebSockets**, enabling real-time status updates under 500ms.

06/2023 – 11/2023

Remote, India

Freelance Software Engineer

Workforce Telemetry & Operations

Real-Time Telemetry & Geofencing Platform

- Created and developed a real-time workforce telemetry platform using React and FastAPI backend, serving **1,000+ active field agents**.
- Built an interactive live map dashboard in **React** using Ant Design to display speeds and alerts, reducing UI load times by 30%.
- Developed an asynchronous location ingestion pipeline utilizing **FastAPI**, **Celery**, and **Redis**, processing **10,000+ concurrent GPS pings with 99.9% uptime**.

SELECTED PROJECTS

MERN FinTech & E-commerce Platform

A collection of full-stack web applications featuring secure user authentication, interactive dashboards, and REST APIs built with React, Node, Express, and MongoDB.

STACK: React.js / Node.js / Express.js / MongoDB / JWT / Redux

Facial Recognition Attendance System

Developed an automated student attendance tracking application integrating Python face detection algorithms (OpenCV and LBPH) with MySQL database.

STACK: Python / OpenCV / Tkinter / MySQL / Pandas

Himalayan Edges Website Development

Launched an e-commerce platform using React.js and Django with PWA capabilities, hosted on AWS, increasing page speed by 30%.

STACK: React.js / Django / SQLite / AWS / PWA

Traveling Portal & Landing Page

A fully responsive, modern front-end landing page for a travel agency built with HTML, CSS, and Bootstrap layouts.

STACK: HTML5 / CSS3 / JavaScript / Bootstrap

Persistent Task Management Dashboard

Programmed a desktop utility for task planning, category filtering, and JSON data persistence, featuring clean MVC architecture patterns.

STACK: Python / Tkinter / JSON / MVC Patterns

Real-Time Telemetry Geofencing Engine

Built a real-time workforce geolocation tracking engine with asynchronous GPS coordinate logs, map visualizations, and alerts via WebSockets.

STACK: React.js / FastAPI / Celery / Redis / PostgreSQL

EDUCATION

Bachelor of Technology (B.Tech.) | Kanpur Institute of Technology, Kanpur, UP

2019 - 2023

CERTIFICATIONS

Advanced MERN Stack Developer Certification
Database Systems & SQL Certification

ACHIEVEMENTS

- Graduated with First Class Honors, maintaining top academic rank.
- Earned a 4-Star Gold Badge in SQL on HackerRank.